

Travaux Pratiques

Semestres 5 et 6

Vision Industrielle

Ressources



Ce sujet est disponible au format électronique sur le site du LEnsE - <https://lense.institutoptique.fr/> dans la rubrique Année / Première Année / Opto-Electronique S5 / TP / Bloc Vision Industrielle.

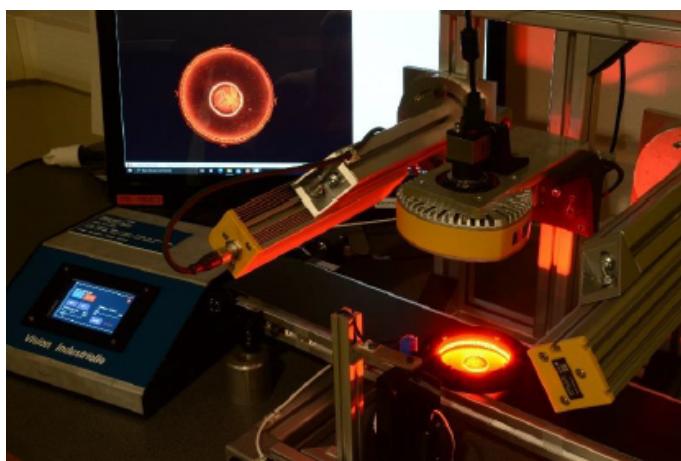


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L'image de la page de garde provient du projet DEPhI Vision Industrielle de 2026. Crédits : Joséphine BECHU, Justine GABRIEL et Paul CHENEAU (Promo 2027).

Vision Industrielle

La **vision industrielle** est une technologie qui permet à des machines d'**analyser automatiquement des scènes** pour **contrôler, guider ou inspecter** des objets sur des processus de production. Elle repose sur l'utilisation de **caméras**, d'**optique**, d'**éclairages** spécifiques (ou contraints), de **capteurs** et d'algorithmes de **traitement d'image**.



Elle a pour but de **prendre des décisions automatiques** (ou aider l'être humain dans sa prise de décision) vis-à-vis d'un (ou plusieurs) objet(s) dans une scène spécifique : détecter des défauts ou des irrégularités, compter ou trier..., en rejetant ou validant automatiquement des produits, tout en assurant une constance de la qualité et de la répétabilité des opérations.

Ressources

Ce bloc de travaux pratiques utilise un **banc de vision industrielle** avec une lampe de type Effi-Ring RGB, une caméra Basler et une interface développée en **Python** (PyQt6) et qui utilise des fonctionnalités de la bibliothèque **OpenCV**.

Les documentations de la caméra et de l'éclairage sont disponibles aux adresses suivantes :

- Basler **a2A 1920 - 160ucBAS** : <https://docs.baslerweb.com/a2a1920-160ucbas#specifications>
- **Effi-Ring** : <https://www.effilux.com/fr/produits/annulaire/effi-ring>

Liste des ressources présentes dans ce document

- Camera BASLER a2A1920-uc/umBAS (Résumé)
- Source EFFI-Ring - Spectre et données / Version RGB
- Cubes de couleur - Réflectance
- Contraste d'un objectif / FTM
- Rappels sur les caméra CMOS
- Rappels sur le traitement d'images

Camera BASLER a2A1920-uc/umBAS

La documentation complète se trouve sur le site du fabricant - Basler

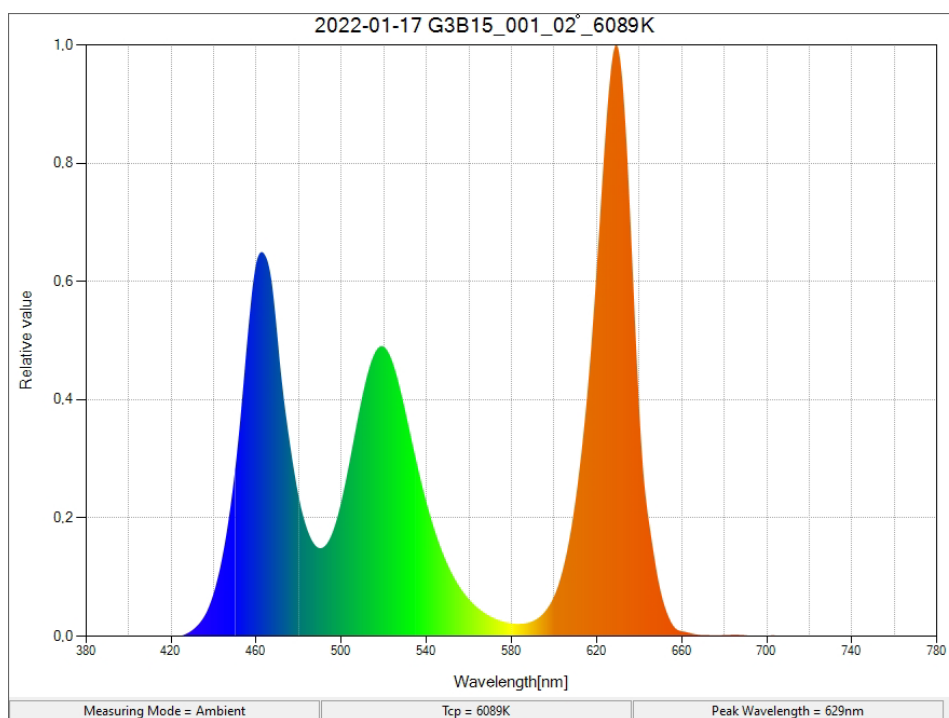
Marque	Basler
Modèle	a2A1920-160ucBAS
Résolution	1920 px x 1200 px
Taille pixel	3.45 x 3.45 μm^2
Profondeur	12 bits
Efficacité quantique	62.22 %
Gain (1/K)	2.652 e-/DN
Capacité de saturation	10492 e-
Capacité de saturation	16862 p

Source EFFI-Ring - Spectre et données / Version RGB

La documentation complète se trouve sur le site du fabricant - Effilux

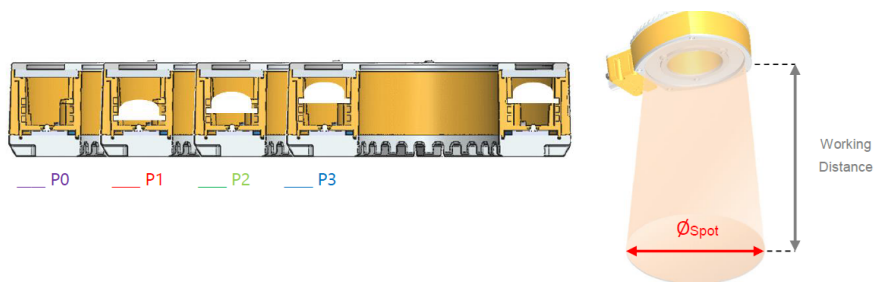
Spectre

Obtenu à l'aide d'un spectromètre



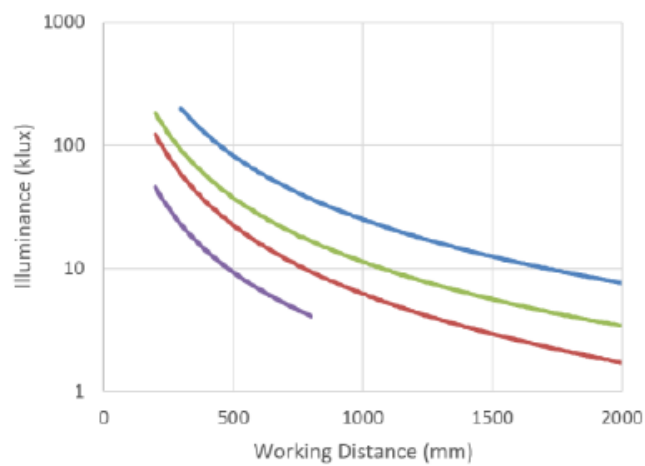
Taille du spot et éclairage en fonction de la distance de travail

Données provenant de la documentation technique.



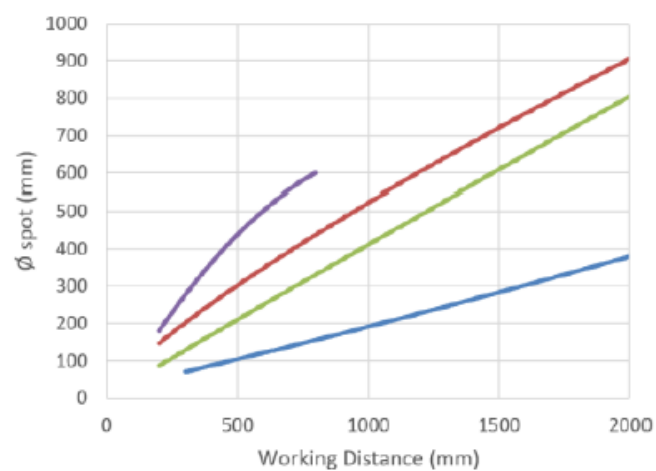
Illumination¹ vs. Working distance

Semi-Diffuse



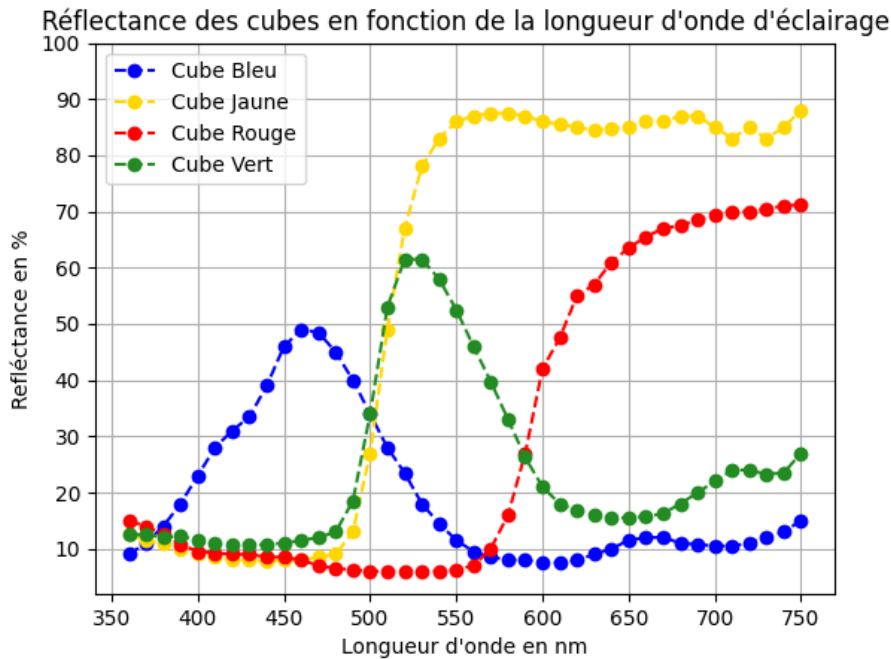
$\varnothing_{spot}^3$ vs. Working distance

Semi-Diffuse



Cubes de couleur - Réflectance

Les **mesures de réflectance** ont été acquises par un **spectro-colorimètre** de type *KONICA MINOLTA CM2600d* (méthode proposée en TP de photométrie en 2ème année du cursus ingénieur)



Contraste d'un objectif / FTM

La **Fonction de Transfert de Modulation** (FTM) (*Modulation Transfer Function* en anglais - *MTF*), permet de caractériser la capacité d'un système optique à restituer les détails d'un objet. Elle décrit comment le **contraste d'un motif périodique** est transmis par le système **en fonction de la fréquence spatiale** (c'est-à-dire de la finesse des détails observés).

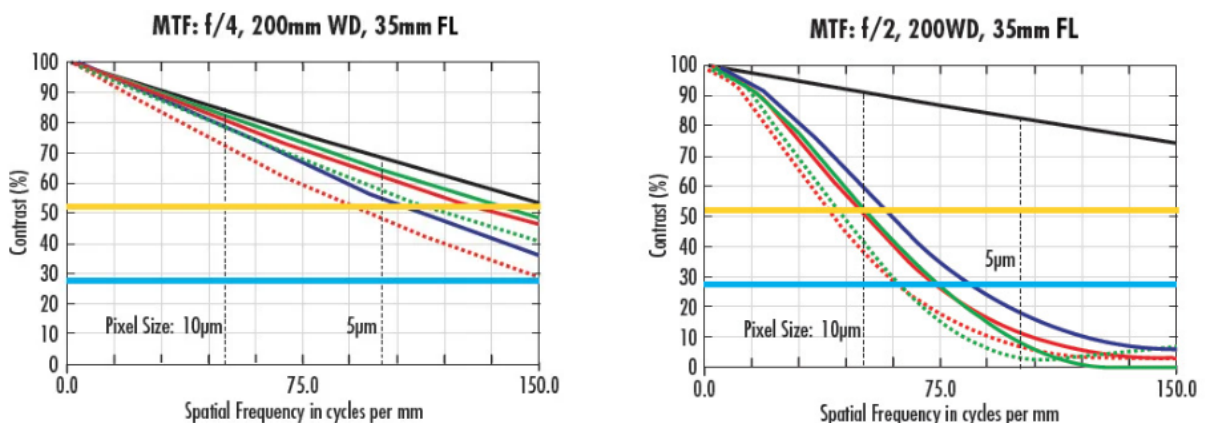
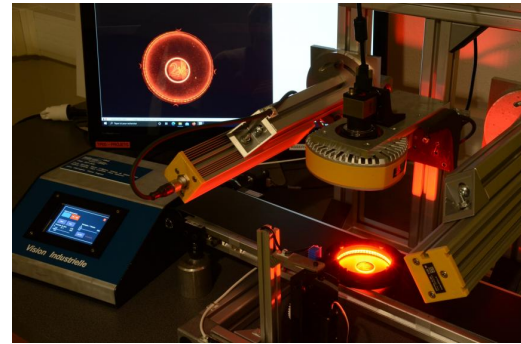


FIGURE 1 – Courbes FTM pour un objectif de 35 mm à la même WD et à différentes ouvertures : f/4 (a) et f/2 (b). *Edmund Optics*

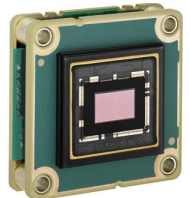
SC 19 – Machine Vision

Cameras and Interfaces

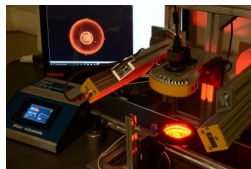
Julien VILLEMEJANE



IDS Sensor

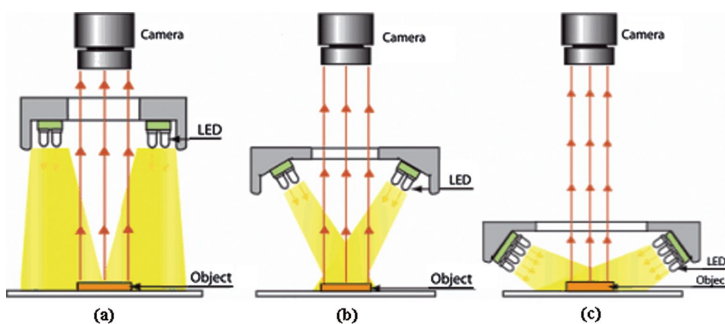


Basler Sensor / Mouser



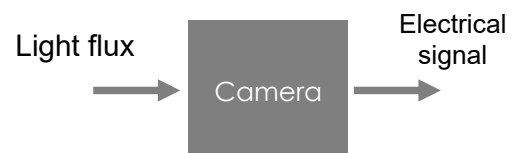
SC19 – Cameras and Interfaces

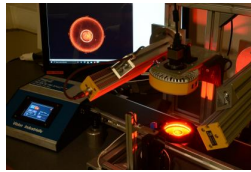
Camera in a machine vision chain



Camera

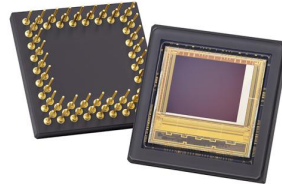
Device that transforms a **light flux** into a **measurable electrical signal**





SC19 – Cameras and Interfaces

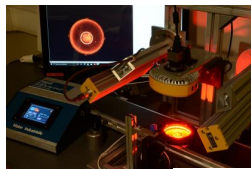
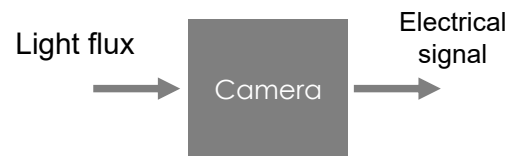
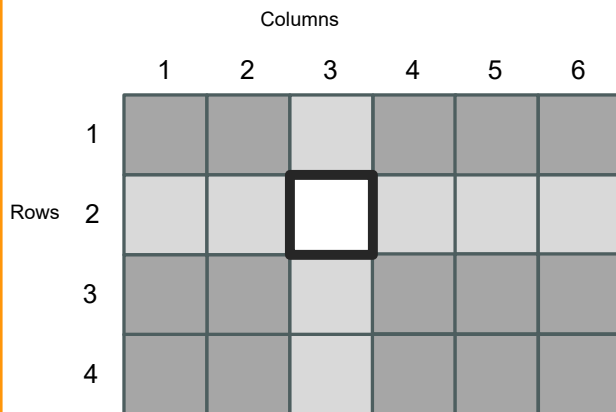
Camera / Array of small sensors



Camera

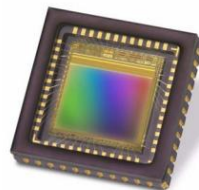
Device that transforms a **light flux** into a **measurable electrical signal**

<https://imaging.teledyne-e2v.com/products/2d-cmos-image-sensors/onyxmax/>

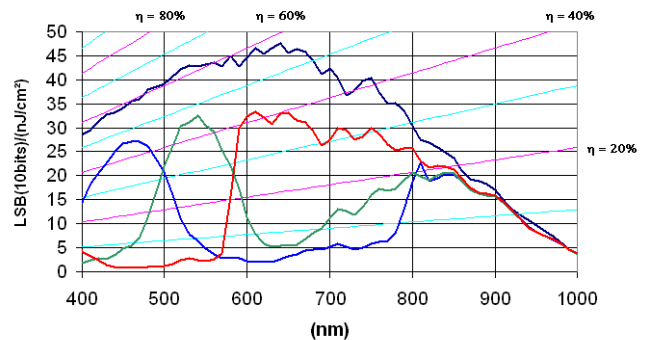
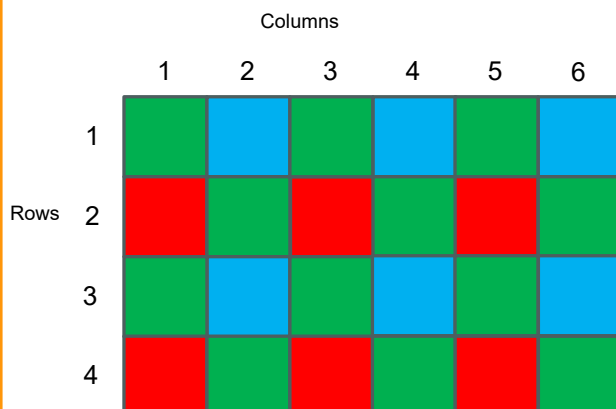


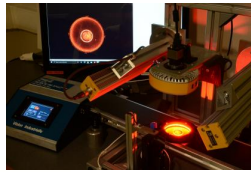
SC19 – Cameras and Interfaces

Camera / Bayer filter for color sensors



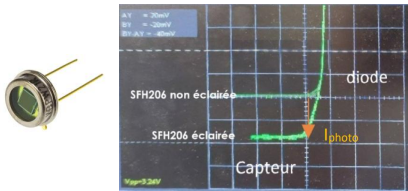
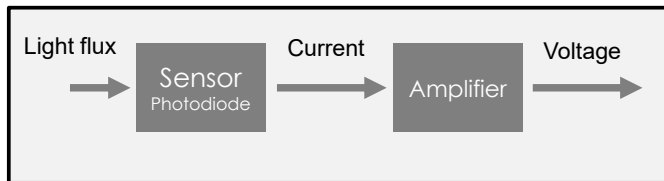
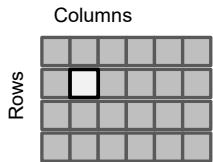
e2v sensor EV76C560ACT





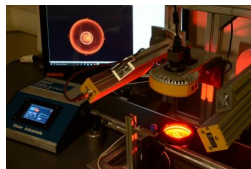
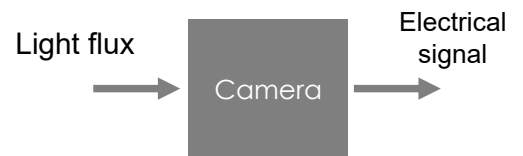
SC19 – Cameras and Interfaces

Camera / Inside a pixel



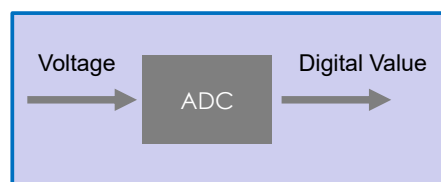
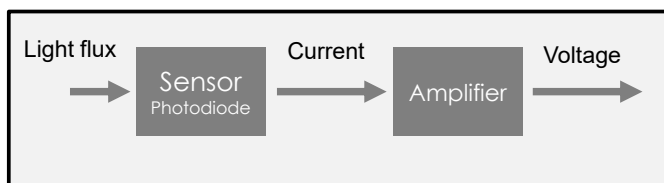
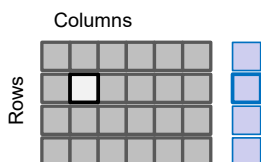
Camera

Device that transforms a **light flux** into a **measurable electrical signal**



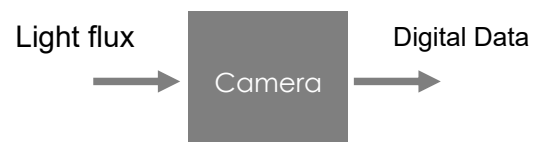
SC19 – Cameras and Interfaces

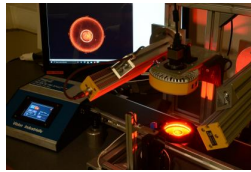
Camera / From analog to digital signal



Digital Camera

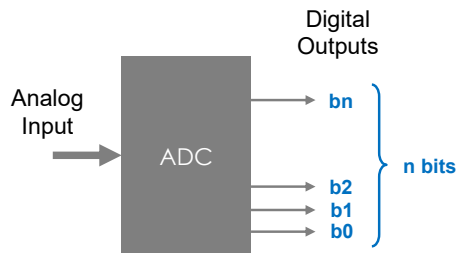
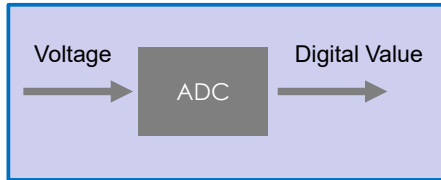
Device that transforms an array of **light flux sensors** into **digital data** called pixels





SC19 – Cameras and Interfaces

How an Analog to Digital Converter works ?

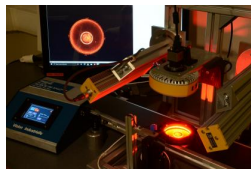
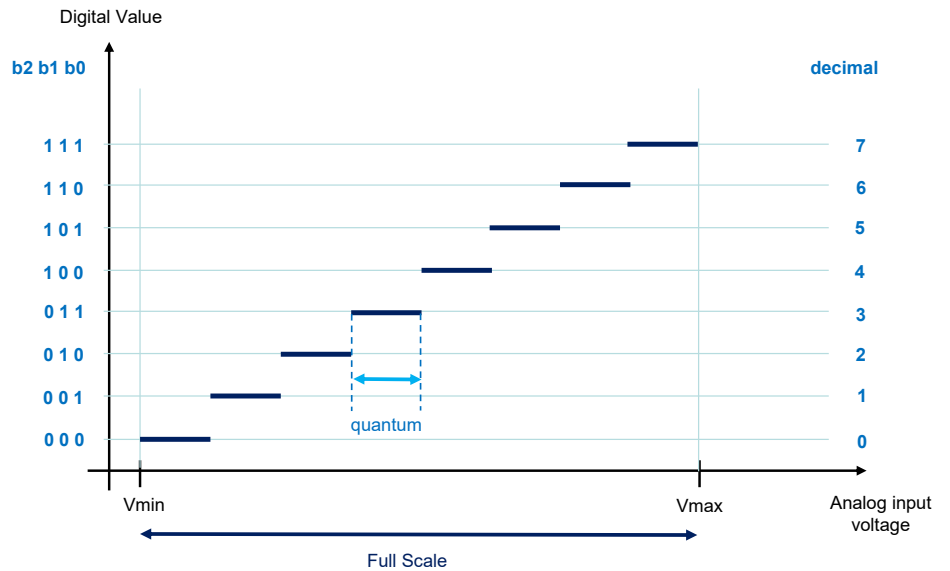


Each bit can have one of two values: **0** or **1**.

The **number of different values** that can be represented by **n bits** is 2^n .

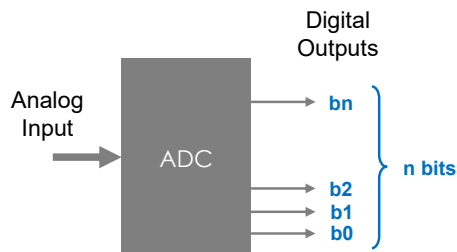
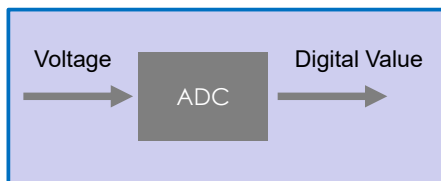
Example for $n = 3$ bits

Quantization



SC19 – Cameras and Interfaces

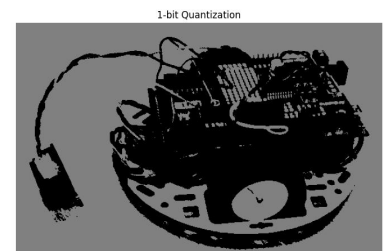
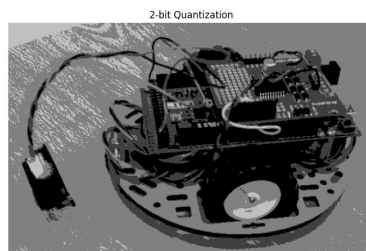
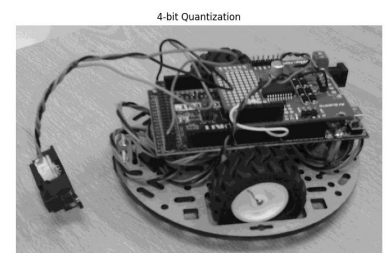
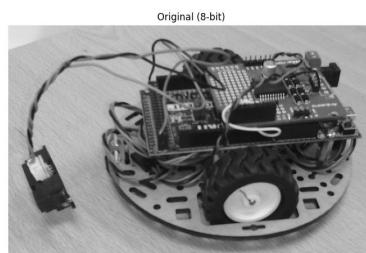
Sampling and quantization of an image

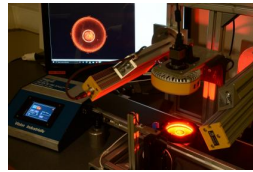


Each bit can have one of two values: **0** or **1**.

The **number of different values** that can be represented by **n bits** is 2^n .

Quantization





SC19 – Cameras and Interfaces

Sampling and quantization of an image



Sampling

Barcode to decode

Area of sampling

<https://barcode-coder.com/fr/specification-ean-13-102.html>



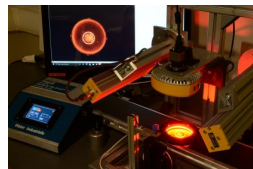
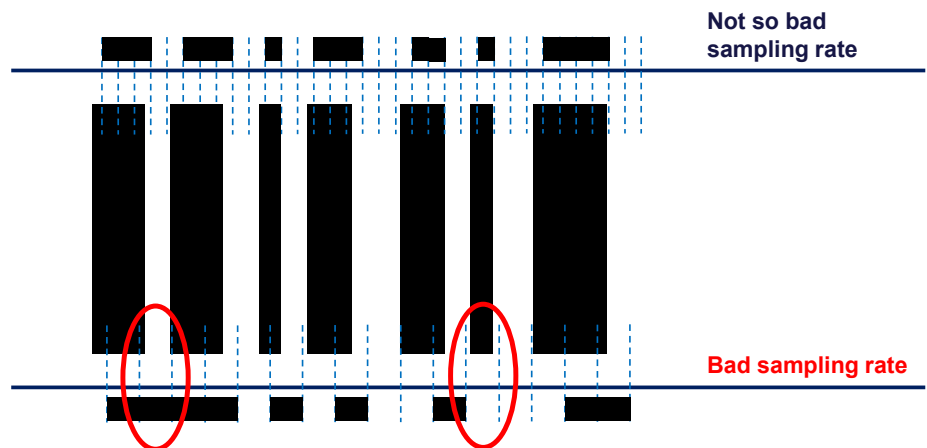
LEnSE 2024

Sampling theorem

Nyquist–Shannon sampling theorem

The sampling frequency must be equal to or **greater than twice** the frequency associated with the finest detail in the image (edges).

With a grid spacing of d , a periodic component with a period higher than $2.d$ can be reconstructed.

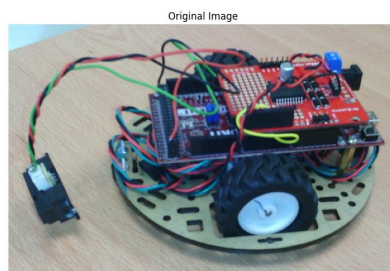


SC19 – Cameras and Interfaces

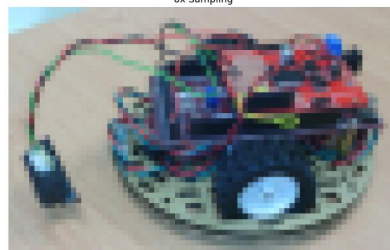
Sampling and quantization of an image



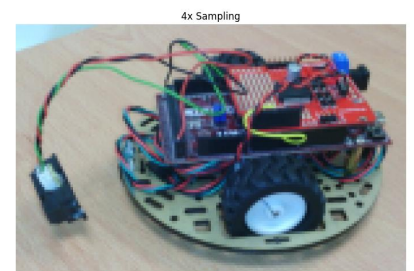
Sampling



Original Image



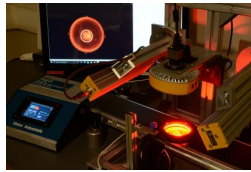
8x Sampling



4x Sampling

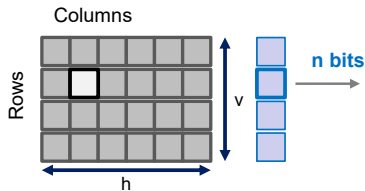


16x Sampling



SC19 – Cameras and Interfaces

Quantity of data per image

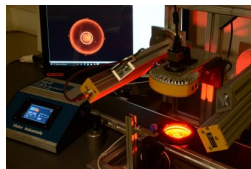


$$\text{Nb of pixels} = h \times v$$

Each pixel is converted into **n bits**.

Each image has a total amount of binary data :

$$\text{Nb of data (bits)} = \text{Nb of pixels} \times n$$



SC19 – Cameras and Interfaces

Frame Rate

Each image has a total amount of binary data :

$$\text{Nb of data (bits)} = \text{Nb of pixels} \times n$$

The amount of data per second :

$$\text{Nb of data per s (bits/s)} = \text{Nb of data (bits)} \times \text{FPS}$$

Frame rate

Number of individual frames
captured **per second** by a device

Expressed in frames per second
(fps)

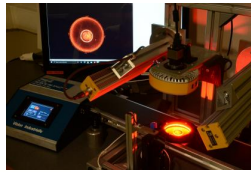
*Higher framerates result
in smoother motion in
video footage*

Example for a 4k camera in 12 bits @ 30 fps :

$$\text{Nb of data (bits)} = 3840 \times 2160 \times 12 = 99\,532\,800 \text{ bits}$$

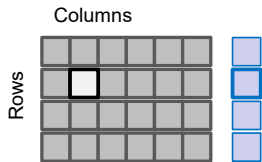
$$\text{Nb of data per s (bits/s)} = 99\,532\,800 \times 30 = 2,9 \text{ billions of bits / s} = 2,78 \text{ Gbit/s}$$

In 2024, the transfer rate of a home router (optical fiber) is theoretically 8 Gbit/s (Free telecom - France)



SC19 – Cameras and Interfaces

Black level : an offset to compensate electronic defaults

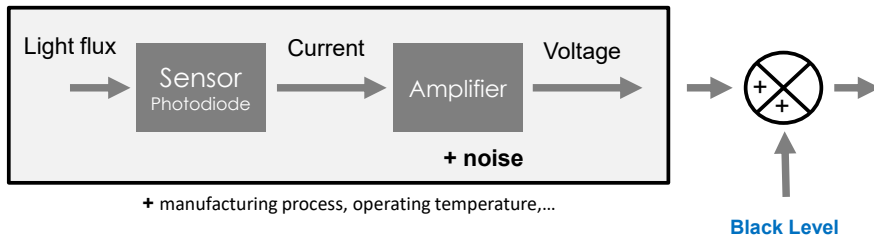


Dark Current

Response of the sensor to **complete darkness**

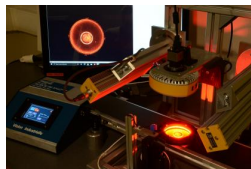
Black Level

Change the **overall brightness** of an image.



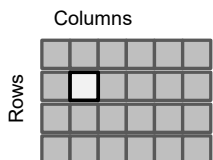
Adjusting the camera's black level will result in **an offset to the pixel's gray values** output by the camera.

Due to **various physical and electronic factors**, the sensor's output is never zero, even in the complete absence of light



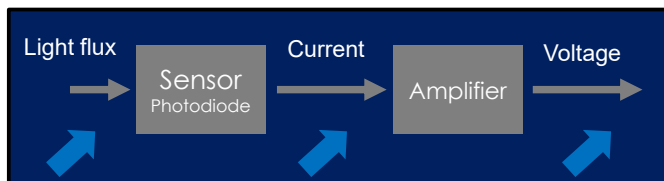
SC19 – Cameras and Interfaces

Exposure Time

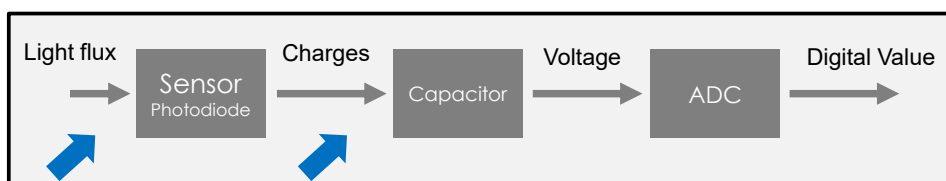


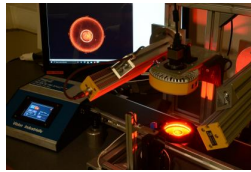
Exposure Time

Duration for which the **camera's sensor is exposed to light**, when capturing an image.



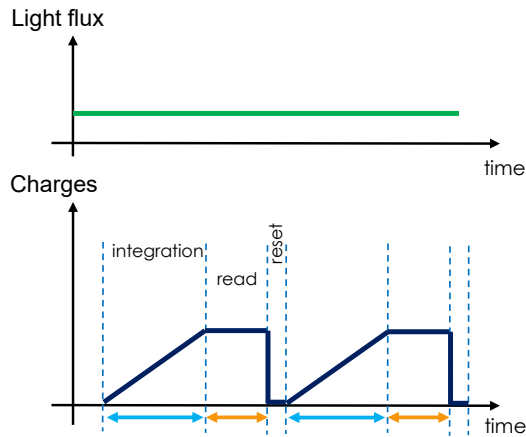
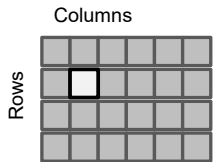
This parameter determines the amount of light collected.





SC19 – Cameras and Interfaces

Exposure Time

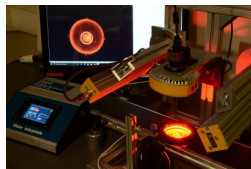
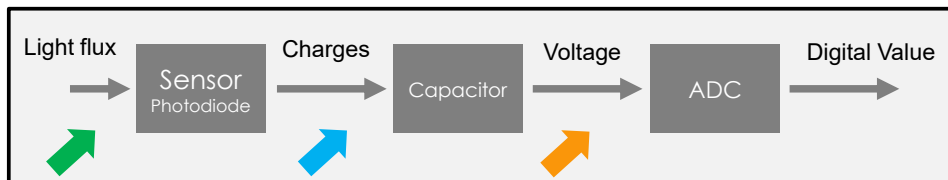


Exposure Time

Duration for which the **camera's sensor is exposed to light**, when capturing an image.

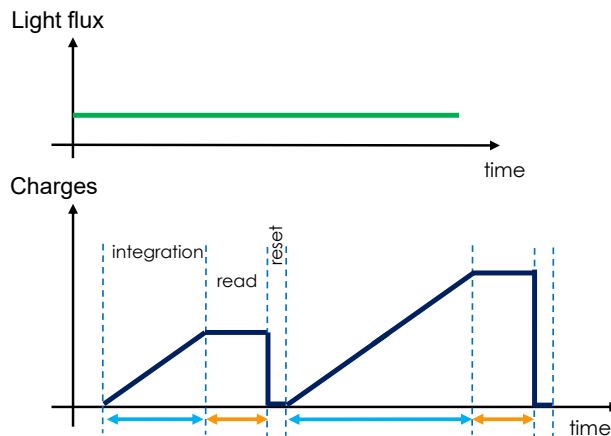
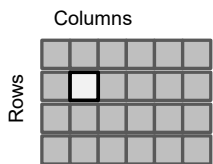
This parameter determines the amount of light collected.

i.e. the amount of collected charges coming from the sensor stored in a capacitor



SC19 – Cameras and Interfaces

Exposure Time

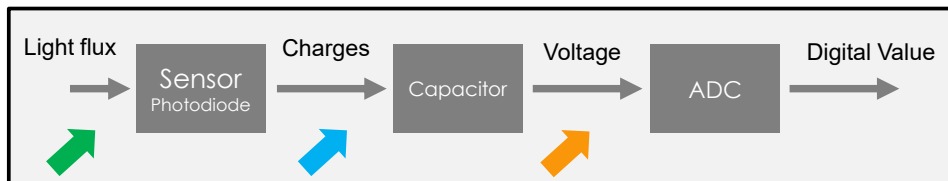


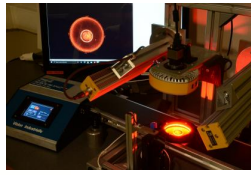
Exposure Time

Duration for which the **camera's sensor is exposed to light**, when capturing an image.

This parameter determines the amount of light collected.

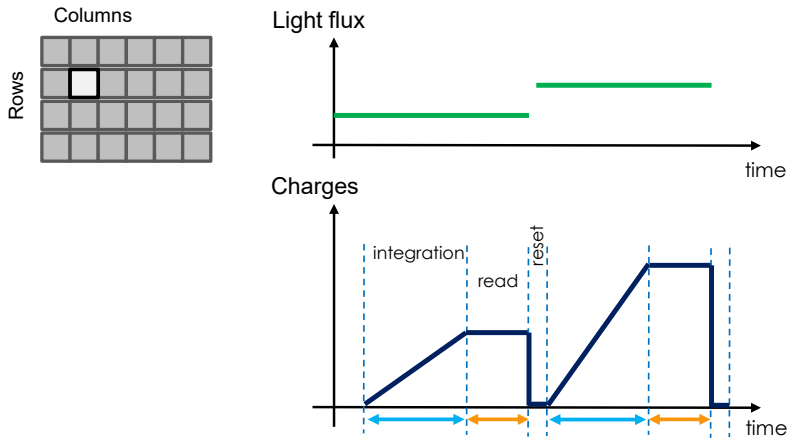
i.e. the amount of collected charges coming from the sensor stored in a capacitor





SC19 – Cameras and Interfaces

Exposure Time

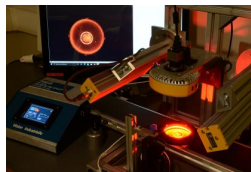
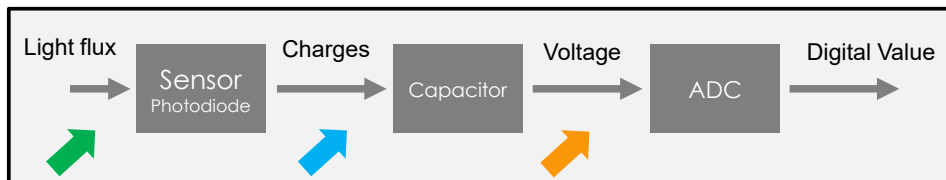


Exposure Time

Duration for which the **camera's sensor is exposed to light**, when capturing an image.

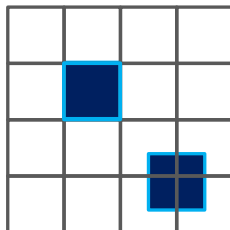
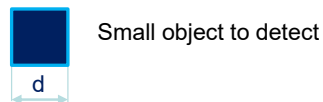
This parameter determines the amount of light collected.

i.e. the amount of collected charges coming from the sensor stored in a capacitor

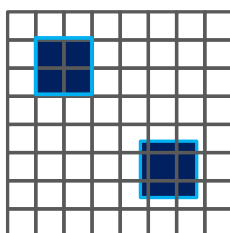


SC19 – Cameras and Interfaces

Spatial Resolution



$$P = d$$



Security factor S

$$P = \frac{d}{S}$$

Spatial resolution / P

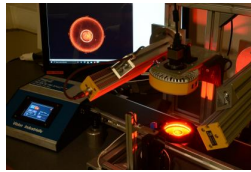
Distance observed by a single **pixel** in a given direction

This security factor is due to the Nyquist-Shanon theorem.

And $S \geq 2$

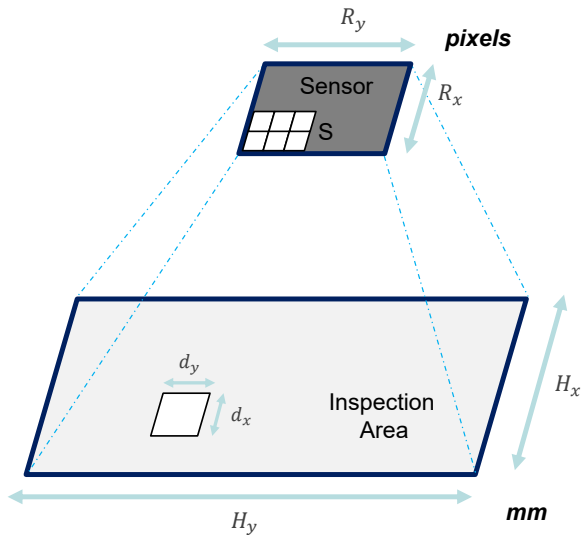
*To verify if the spatial resolution is good enough, **calibration target** can be used. (Foucault)*





SC19 – Cameras and Interfaces

Resolution of the sensor



Spatial resolution / P

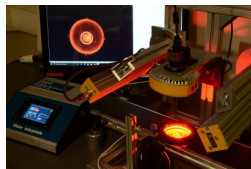
Distance observed by a single
pixel in a given direction

$$P = \frac{d}{S}$$

Sensor resolution (pixels)

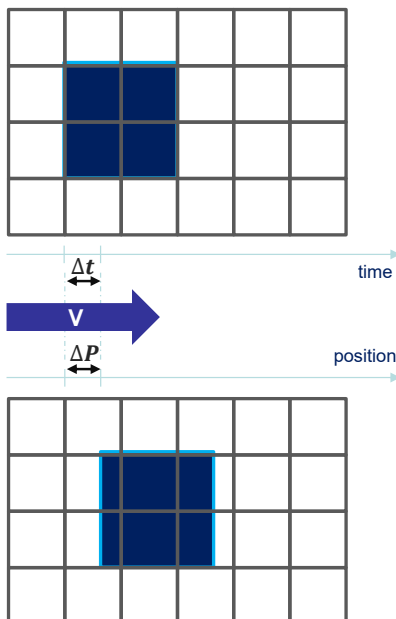
$$R = \frac{H}{P} = \frac{S \times H}{d}$$

H (mm)	→	R (px)
d (mm)	→	S (px)
P (mm)	→	1 (px)



SC19 – Cameras and Interfaces

Motion, sharp image and maximum exposure time



V : motion speed (mm/s)

Spatial resolution / P

Distance observed by a single
pixel in a given direction

$$P = \frac{d}{S}$$

Displacement

$P \times \Delta P$ (mm) → Δt (s)

Time

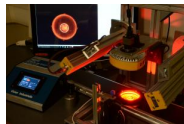
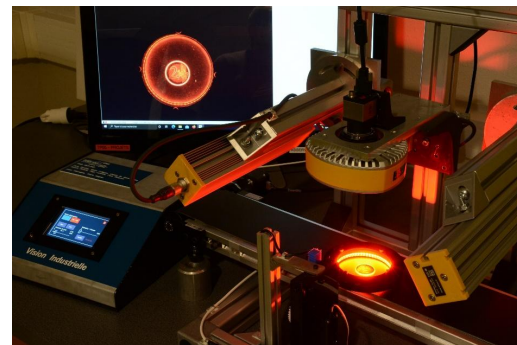
$$\Delta t = \frac{P \times \Delta P}{V}$$

Motion blur perception threshold
to obtain a sharp image
is between

1/2 and 1/5 of a pixel

Traitement d'image

Pré-traitement / Segmentation / Classification



Traitement d'images

Objectif



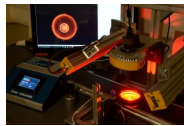
Image brute 'RAW' / Caméra

- **Bruitée**
- Mauvais contraste
- Eclairage non uniforme
- ...



Image souhaitée / Contours bien définis

- Zones homogènes
- Transitions nettes



Traitement d'images

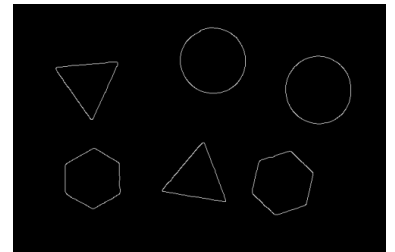
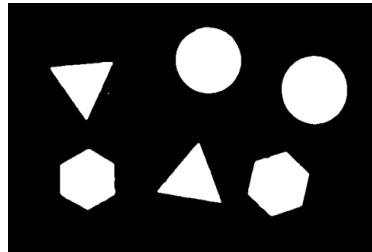
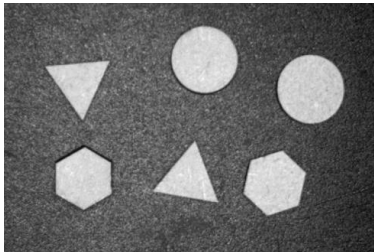


Image brute 'RAW' / Caméra

- **Bruitée**
- Mauvais contraste
- Eclairage non uniforme
- ...

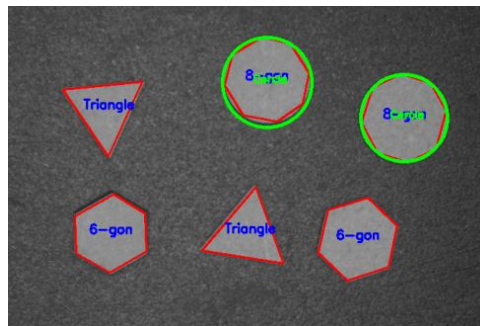
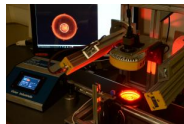


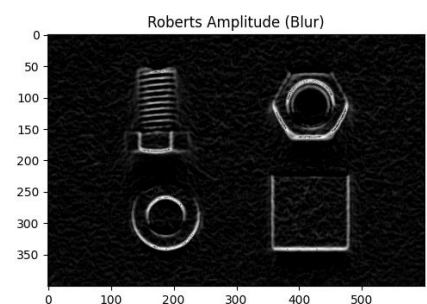
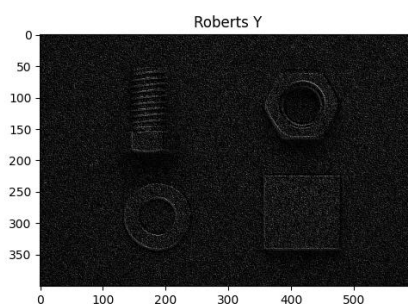
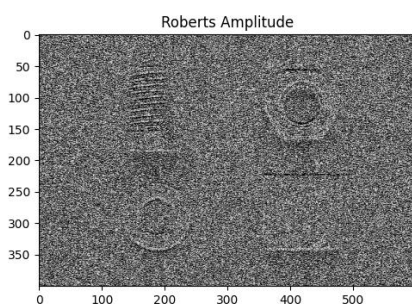
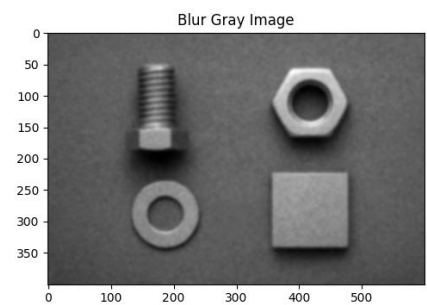
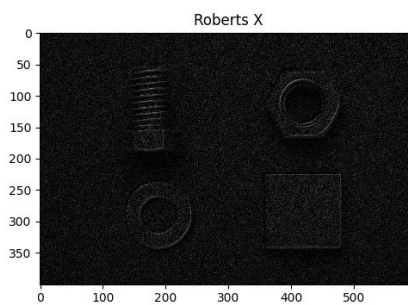
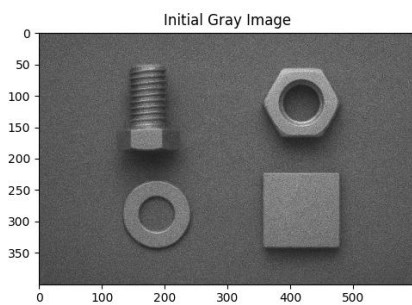
Image souhaitée / Contours bien définis

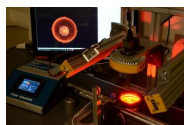
- Zones homogènes
- Transitions nettes



Traitement d'images

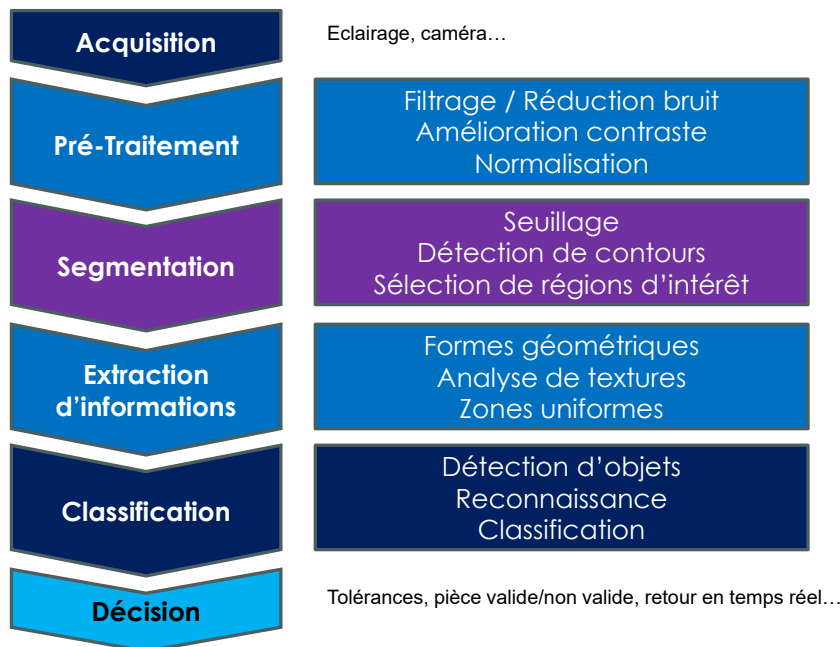
Exemple industriel





Traitement d'images

Objectif

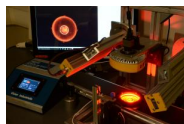


Améliorer la clarté de l'image / réduire les informations indésirables
Faire ressortir les caractéristiques d'intérêt
Standardiser l'échelle ou l'intensité de l'image

Isoler les objets de la région d'intérêt (ROI)
Séparer les objets de l'arrière-plan
Identifier les limites et les contours
Se concentrer uniquement sur les parties pertinentes de l'image

Extraire des données (taille, forme, position...)
Reconnaître des formes, des symboles ou des points d'intérêt

Identifier et nommer des objets
Vérifier si les données mesurées sont en accord avec un cahier des charges
Catégoriser des objets dans des groupes spécifiques



Traitement d'images

OpenCV

Open Source Computer Vision

Une bibliothèque de **traitement d'images**
et de **Machine learning**

Développés sur de multiple environnement,
comme Python, C++, Java, and MATLAB

Traitement d'images	Filtrage, detection de contours, transformations...
Reconnaissance	Détection d'objets dans des images et des vidéos
Algorithmes Vidéo	Suivi de mouvement, Reconstruction 3D...
Machine Learning	Classification d'images, Reconnaissance de formes

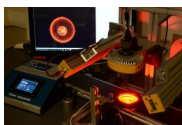


<https://opencv.org>



OpenCV 4.5.0 and higher versions are licensed under the [Apache 2 License](#).

OpenCV 4.4.0 and lower versions, including OpenCV 3.x, OpenCV 2.x, and OpenCV 1.x, are licensed under the [3-clause BSD license](#).



Traitement d'images

Images numériques

Image continue

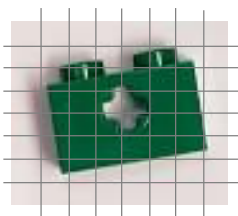
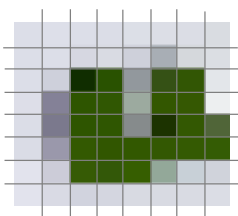


Image numérique : projection sur une matrice d'une image continue



8 x 8 grid



16 x 16 grid

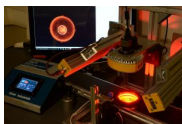


32 x 32 grid

Image numérique

Représentation d'une image
sous forme numérique

Pour être **sauvegardée, traitée**
et **affichée** par des ordinateurs
ou des systems numériques.

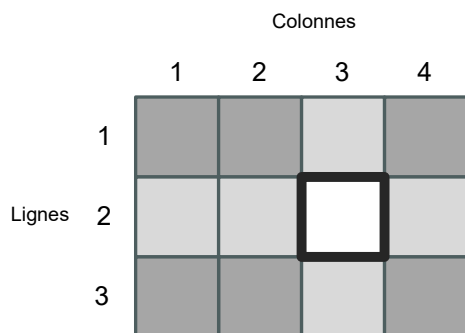


Traitement d'images

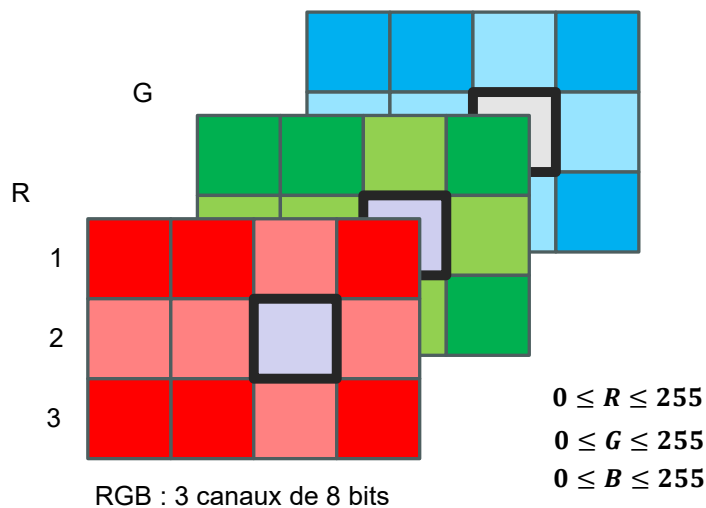
Images numériques / Gris ou RGB

Gris

Nb de pixels = $h \times v$



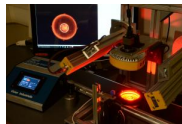
Chaque pixel est converti sur **n bits**.



RGB : 3 canaux de 8 bits



R=200, G=100, B=50

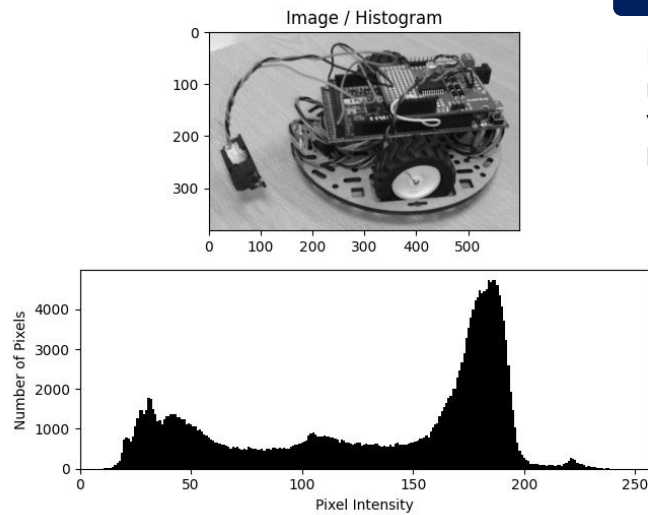


Traitement d'images

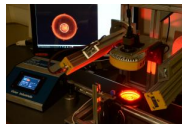
Filtrage par TF

Acquisition

Histogramme



Représentation graphique montrant la distribution des valeurs de niveaux de gris des pixels de l'image

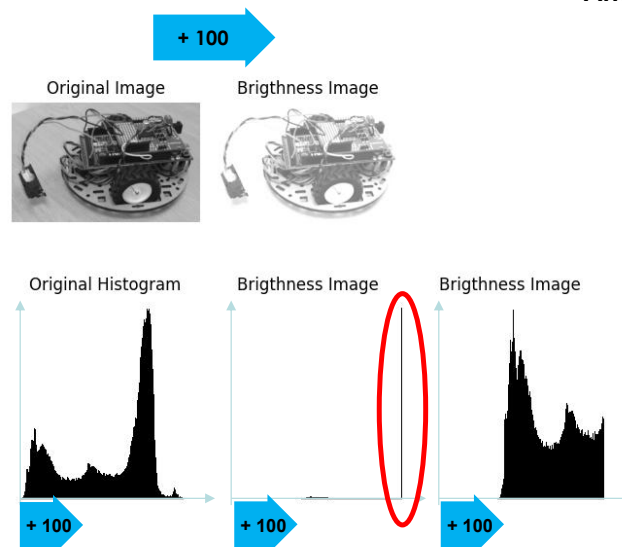


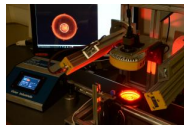
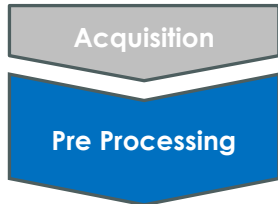
Traitement d'images

Amélioration de l'image

Acquisition

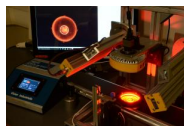
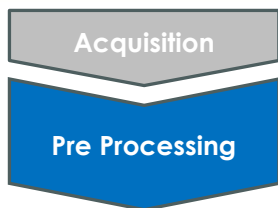
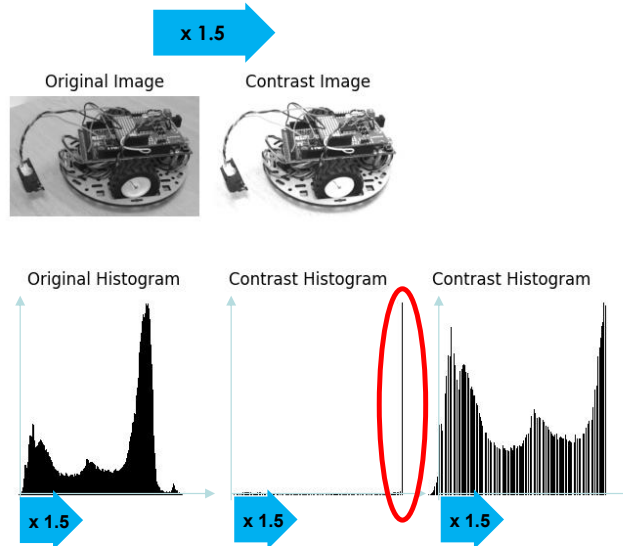
Pre Processing





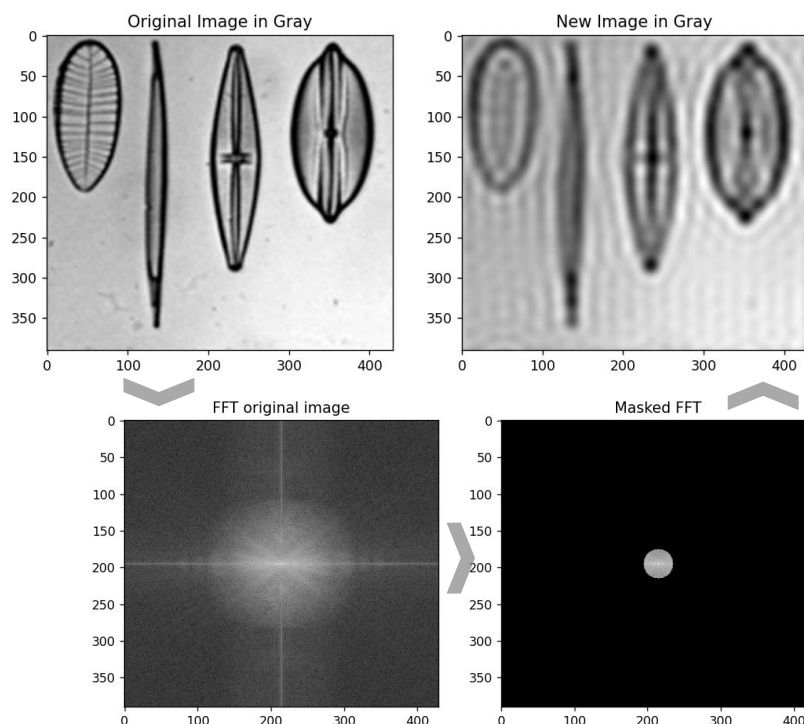
Traitement d'images

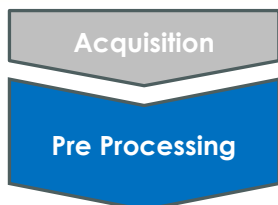
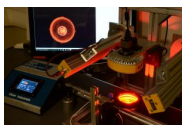
Amélioration de l'image



Traitement d'images

Filtrage par TF





kernel

-1	0	-2
1	5	1
-2	0	-1

original image

5	8	4	2	3	1	5
9	5	1	8	7	6	2
5	7	1	5	6	8	7
5	8	2	8	4	3	3
5	6	6	7	2	5	1

Traitement d'images

Filtrage / Convolution

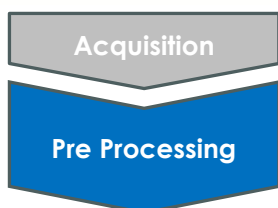
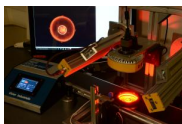
5	8	4	2	3	1	5
9	5	1	8 x -1	7 x 0	6 x -2	2
5	7	1	5 x 1	6 x 5	8 x 1	7
5	8	2	8 x -2	4 x 0	3 x -1	3
5	6	6	7	2	5	1

filtered image

$$R = -8 + 0 - 12 + 5 + 30 + 8 - 16 + 0 - 3$$

$$R = 4$$

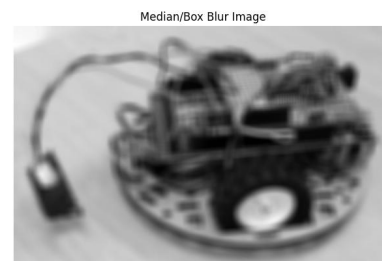
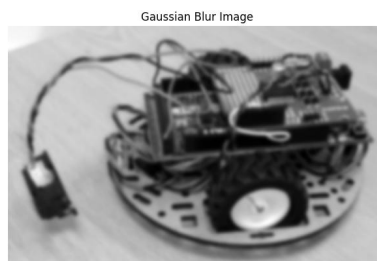
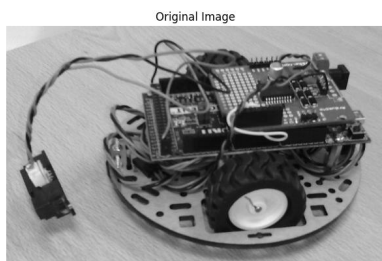
				4		



Traitement d'images

Filtrage / Convolution

Suppression de détails insignifiants

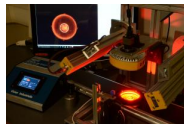


1	4	7	4	1
4	16	26	16	4
7	26	41	26	7
4	16	26	16	4
1	4	7	4	1

Gaussian Kernel
(x 1/273)

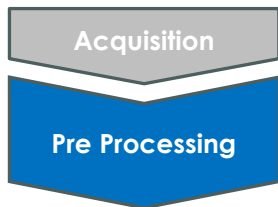
Mean Kernel (x 1/(N*M))

1/9	1/9	1/9
1/9	1/9	1/9
1/9	1/9	1/9

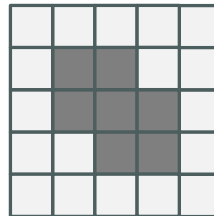
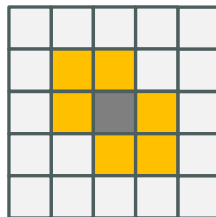


Traitement d'images

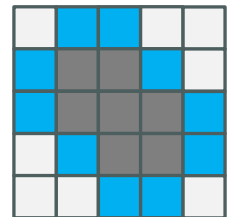
Erosion / Dilatation



■ Pixels originaux
■ Pixels retirés



■ Pixels ajoutés



Erosion

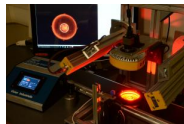
Réduire le premier plan en retirant progressivement les pixels le long des contours des objets

Dilatation

Étendre le premier plan en ajoutant des pixels le long des contours des objets

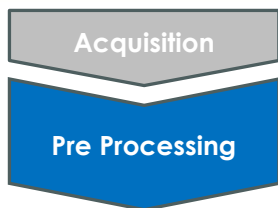
kernel

0	1	0
1	1	1
0	1	0



Traitement d'images

Erosion / Dilatation



Eroded Image



Original Image



Dilated Image



Erosion

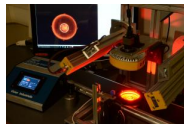
Réduire le premier plan en retirant progressivement les pixels le long des contours des objets

Dilatation

Étendre le premier plan en ajoutant des pixels le long des contours des objets

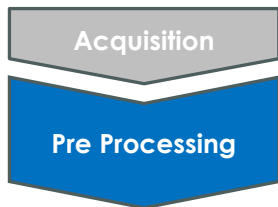
kernel

0	1	0
1	1	1
0	1	0



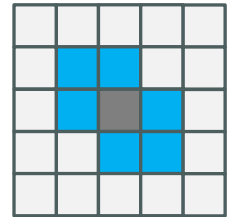
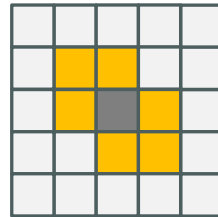
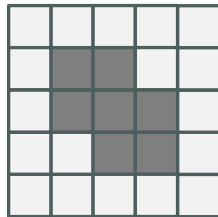
Traitement d'images

Ouverture / Fermeture



Original pixels
Removed pixels

Added pixels



Ouverture

Fermeture

Erosion puis Dilatation

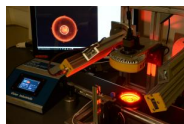
Dilatation puis Erosion

Retire des petits objets

Remplit des petites zones

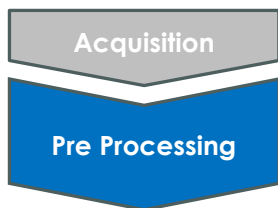
kernel

0	1	0
1	1	1
0	1	0



Traitement d'images

Ouverture / Fermeture



Opening Image

Original Image

Closing Image



Ouverture

Fermeture

Erosion puis Dilatation

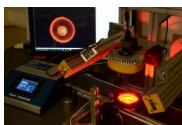
Dilatation puis Erosion

Retire des petits objets

Remplit des petites zones

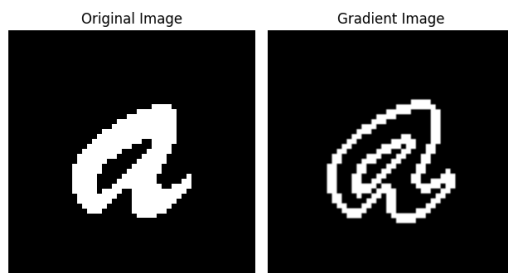
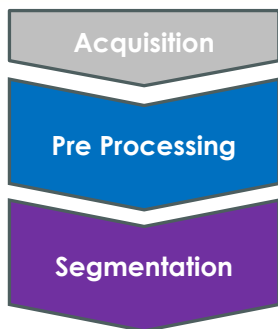
kernel

0	1	0
1	1	1
0	1	0



Traitement d'images

Gradient



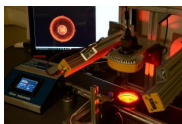
kernel

0	1	0
1	1	1
0	1	0

Gradient

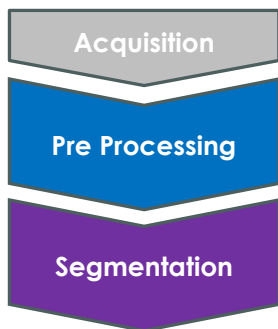
Difference entre une **dilatation** et une **érosion**

Classification des pixels : **scène** (background) ou **objets** (foreground) ?



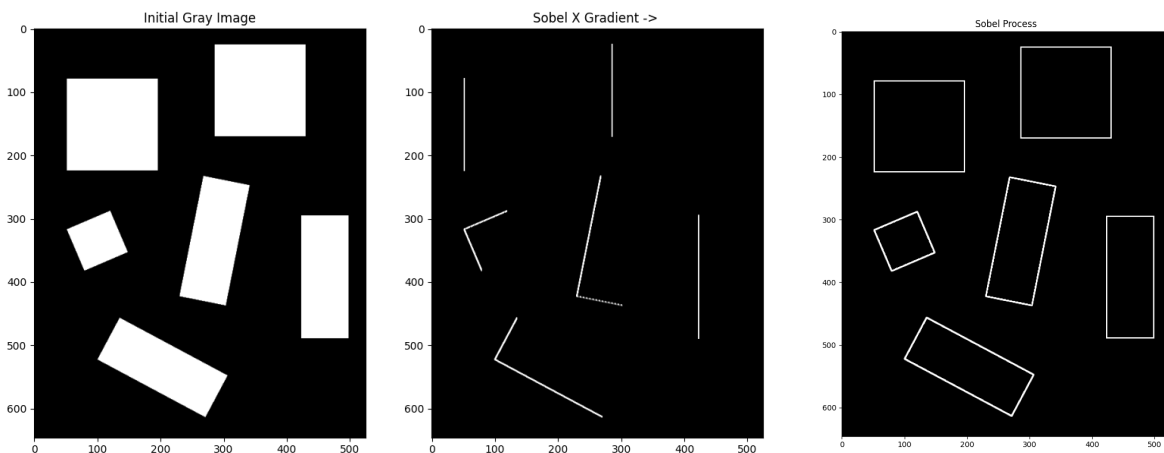
Traitement d'images

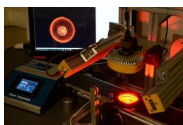
Opérateur de Sobel



kernel

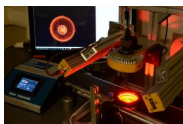
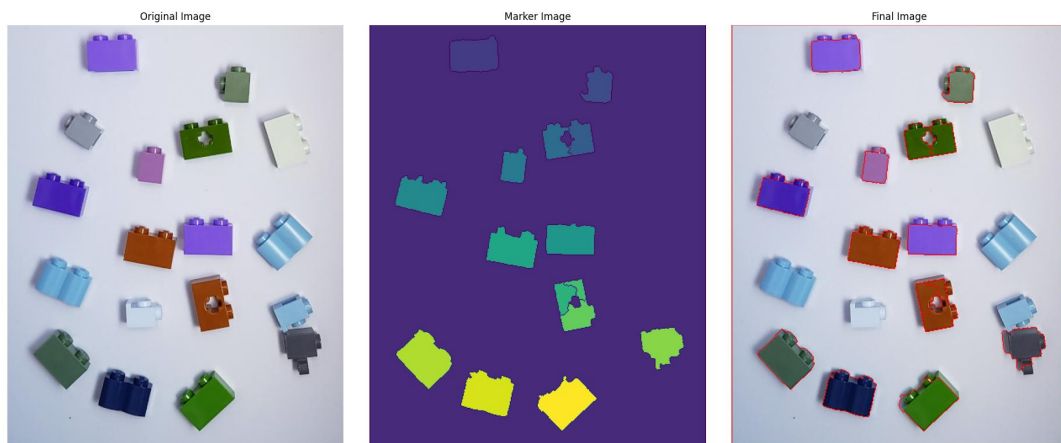
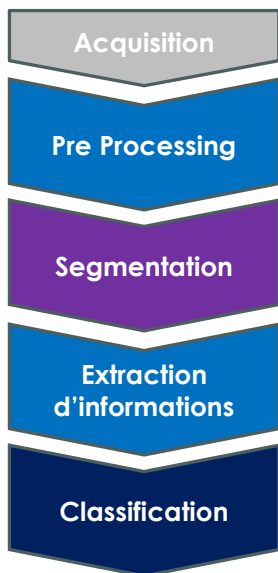
-1	0	1
-2	0	2
-1	0	1





Traitement d'images

Méthode de Watershed



Traitement d'images

Méthode de Watershed

